

decimal code is nothing but decimal 32 ((16<sup>x</sup>2)+(16<sup>x</sup>0)). Hence it is reflected as 32 just below the keyboard in a square box on the left-hand side of the screen (refer Screenshot 2).

This program not only generates the ASCII codes of all the keys present on the keyboard and mouse buttons but also highlights the key being pressed by green colour and the mouse button by violet colour.

In Screenshot 3, the pressing of a mouse button is highlighted in one of the six boxes just below the keyboard and at the same time, the corresponding code is displayed in a square box on the right-hand side of the screen. For example, when you press the left mouse button, code '1' is displayed in the square box on the right-hand side of the screen as shown in Screenshot 3.

Generally, there are eight codes

associated with mouse clicks: 0 for the scroll wheel, 1 for the primary button, 2 for the secondary button, 3 for pressing both the primary and secondary buttons, 4 for clicking the scroll wheel, 5 for clicking the scroll wheel and primary button simultaneously, 6 for clicking the scroll wheel and secondary button, and 7 for clicking all the mouse keys and scroll wheel simultaneously.

To know more about conversions from hexadecimal into ASCII equivalents, you may refer to the jade clip. htm program published in July 2010 issue of EFY.

## Function description

This program uses eleven functions to get data and display the codes of keys pressed. Display of the corresponding key code (ASCII) and highlighting the same on the screen is controlled by

the scan() function. Pressing of 'Shift' key is also scanned, with the alphanumeric display changing as shown in Screenshot 3.

The capsled(), numled() and scrolled() functions help recognise the LED indicators for 'Caps Lock', 'Num Lock' and 'Scr Lock' on the keyboard. To get LED outputs synchronised with the pressing of these keys, ensure that 'Num Lock' key is in 'on' mode when the program is run.

The program scans mouse buttons and traces the position of the mouse on the screen using sniff() and trace() functions. It also scans the rotation of the scroll wheel, senses the direction and displays 'Up' or 'Down' in the box just below 'ScW' box.

**EFY note.** The source code of this article has been included in this month's EFY-CD. ●

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